

MANAV SAMIR BHAVSAR

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EDUCATION

Nirma University

Bachelor of Technology in Computer Science and Engineering

Ahmedabad, India

2023 – 2027 (Expected)

Relevant Coursework: Deep Learning, DBMS, OOP, Cloud Computing, Operating Systems, Machine Learning, AI, Computer Networks, Probability & Statistics, Data Structures & Algorithms.

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy), SQL, C, HTML/CSS, JavaScript (React)

AI & Machine Learning: TensorFlow, PyTorch, Scikit-learn, **LangChain**, **RAG**, **LangGraph**, YOLOv8, LLMs

Data & Analysis: Matplotlib, Seaborn, **Statistical Modeling**, FAISS, ChromaDB, Supabase (pgvector)

Backend & Deployment: Flask, **FastAPI**, **Docker**, Streamlit, Google Cloud Platform (GCP), MLflow, Vercel, Render

EXPERIENCE

Labmentix

Data Science & AI Intern

Remote

Oct 2025 – Present

Spearheading 5+ end-to-end ML/AI projects, streamlining data pipelines and deploying production-ready apps via Streamlit.

- **Agentic AI Travel Planner:** Designed a **RAG-based agent** using LangChain; automated complex itinerary generation from unstructured knowledge bases.
- **EMI Prediction Platform:** Achieved **98% classification accuracy** and **0.96 R^2** by optimizing **XGBoost** models with **MLflow** for experiment tracking.
- **Aerial Object Detection:** Built a real-time detection system using **YOLOv8** and **Transfer Learning** (EfficientNet), reaching **98.8% accuracy**.
- **Stock Forecasting:** Implemented **LSTM** and **SimpleRNN** architectures for time-series forecasting, reducing MSE to 317.93.

PROJECTS

GreenTrace — Carbon Footprint Analyzer | React, FastAPI, Groq LLM, Vercel, Render | Live

- Architected a full-stack cloud platform that profiles **per-cell CO₂ emissions** of Jupyter Notebooks via **Live Trace** (Intel RAPL / psutil hardware profiling) and **Fast Estimate** (AST static analysis) modes — deployed on Vercel + Render.
- Integrated **Groq LLM** to generate ranked AI optimization recommendations (mixed-precision, data loader fixes, hardware accelerators), translating raw energy figures into real-world equivalences (km driven, smartphone charges).
- Conducted original **Green AI research**: benchmarked GAN vs. classical augmentation strategies for a dementia MRI classifier; concluded classical augmentation + transfer learning delivers the optimal accuracy-to-carbon ratio — embedded as a full research article in-app.

Agentic AI Knowledge Chatbot | LangGraph, FAISS, FastAPI

- Created a modular RAG assistant using **LangGraph** for multi-step reasoning, **SentenceTransformers** for embedding, and a **FastAPI** backend with real-time confidence scoring.

Mini RAG Document Q&A System | Jina AI, Groq, Supabase, Streamlit

- Engineered a high-performance RAG pipeline with **semantic retrieval**, **reranking**, **Supabase pgvector** storage, and **Groq LLMs** for near-instant PDF Q&A inference.

STRENGTHS

Professional: Technical Communication, Remote Collaboration, Agile Workflows, Technical Documentation |

Personal: Analytical Problem Solving, Critical Thinking, Adaptability, Deadline-Driven Management

AWARDS & CERTIFICATIONS

- **Ranked 1st**, Ideafy Hackathon (2023)
- **Python for Machine Learning & Data Science Masterclass** (Udemy) – Jose Portilla